

Engineering Standards: Cloud Microservices & API Architecture

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1. Microservices Decomposition Standards

New enterprise services must adhere to domain-driven design (DDD) principles. Core business domains must be decoupled into independent, containerized services interacting via gRPC for high-speed internal calls and REST API Gateways for public interfaces.

2. Zero-Trust Security & API Authentication

Inter-service communications require Mutual TLS (mTLS) authentication. All API endpoints must enforce OAuth2 token validation, rate-limiting, and payload sanitization at the API Gateway layer.

3. Ingestion & Embedding Pipeline Integration

Document ingestion services must parse unstructured content, generate structured JSON metadata events, and publish them to Kafka topics before pushing vector embeddings into ChromaDB or FAISS stores.